Lab sheet-02

```
// TASK 1: Create product array
//================================================
const products = [
    {
        id: 1,
        name: "Headphones",
        price: 2499,
        image: "images/headphones.jpg"
    },
    {
        id: 2,
        name: "Smart Watch",
        price: 3999,
        image: "images/watch.jpg"
    },
    {
        id: 3,
        name: "Bluetooth Speaker",
        price: 1999,
        image: "images/speaker.jpg"
    },
    {
        id: 4,
        name: "Keyboard",
        price: 3499,
        image: "images/keyboard.jpg"
    },
    {
        id: 5,
        name: "Gaming Mouse",
        price: 1499,
        image: "images/mouse.jpg"
    },
    {
        id: 6,
        name: "USB Charger",
        price: 999,
        image: "images/charger.jpg"
    },
    {
        id: 7,
        name: "Laptop Stand",
        price: 1299,
        image: "images/stand.jpg"
    },
    {
        id: 8,
        name: "Webcam",
        price: 2299,
        image: "images/webcam.jpg"
    }
];
// TASK 3: Create cart and use localStorage
//================================================
let cart = JSON.parse(localStorage.getItem("cart")) || [];
function saveCart() {
    localStorage.setItem("cart", JSON.stringify(cart));
}
```

```
// TASK 8: Update cart item count
// ================================================
function updateCartCount() {

    const cartCount = document.getElementById("cart-count");

    if (cartCount) {

        const count = cart.reduce(function(total, item) {
            return total + item.quantity;
        }, 0);

        cartCount.textContent = count;
    }
}

// TASK 2: Dynamically display products
// ================================================
function displayProducts() {
    const productList = document.getElementById("product-list");
    if (!productList) return;
    productList.innerHTML = "";
    products.forEach(function(product) {
        const card = document.createElement("div");
        card.innerHTML = `
            <img src="${product.image}" width="150">
            <h3>${product.name}</h3>
            <p>₹${product.price}</p>
            <button onclick="addToCart(${product.id})">
                Add to Cart
            </button>       `;
        productList.appendChild(card);
    });
}
// TASK 3: Add product to cart
// ================================================
function addToCart(id) {
    const product = products.find(function(product) {
        return product.id === id;
    });
    const existingItem = cart.find(function(item) {
        return item.id === id;
    });
    if (existingItem) {
        existingItem.quantity++;
    } else {
        cart.push({
            id: product.id,
            name: product.name,
            price: product.price,
            image: product.image,
            quantity: 1
        });
    }
   saveCart();
    updateCartCount();
    alert("Product added to cart!");
}
```

```
// TASK 4: Display cart items
// ================================================
function displayCart() {
    const cartItems = document.getElementById("cart-items");
    if (!cartItems) return;
    cartItems.innerHTML = "";
    if (cart.length === 0) {
        cartItems.innerHTML = "<p>Your cart is empty.</p>";
       calculateTotal();
        return;
    }
    cart.forEach(function(item) {
        const div = document.createElement("div");
        div.innerHTML = `
            <h3>${item.name}</h3>
            <p>Price: ₹${item.price}</p>
            <label>
                Quantity:
                <input
                    type="number"
                    min="1"
                    value="${item.quantity}"
                    onchange="changeQuantity(${item.id}, this.value)"
                >
            </label>

            <button onclick="removeFromCart(${item.id})">
                Remove
            </button>
            <hr>
        cartItems.appendChild(div);
    });
    calculateTotal();
}
// TASK 5: Change quantity
// ================================================
function changeQuantity(id, quantity) {
    const item = cart.find(function(item) {
        return item.id === id;
    });
    if (item) {

        item.quantity = Number(quantity);

        if (item.quantity < 1) {
            item.quantity = 1;
        }
    }
    saveCart();
    displayCart();
    updateCartCount();
}
// TASK 6: Remove item from cart
// ================================================
function removeFromCart(id) {
    cart = cart.filter(function(item) {
        return item.id !== id;
    });
    saveCart();
```

```
    displayCart();
    updateCartCount();
}
// TASK 5: Calculate grand total using reduce()
// ================================================
function calculateTotal() {
    const totalElement = document.getElementById("cart-total");

    if (!totalElement) return;

    const total = cart.reduce(function(sum, item) {

        return sum + item.price * item.quantity;

    }, 0);

    totalElement.textContent = "₹" + total;
}

// TASK 7: Checkout form validation
// ================================================
function setupCheckout() {
    const form = document.getElementById("checkout-form");
    if (!form) return;
    form.addEventListener("submit", function(event) {
        event.preventDefault();
        const name = document.getElementById("name").value.trim();
        const address =
            document.getElementById("address").value.trim();
        const pincode =
            document.getElementById("pincode").value.trim();
        const phone =
            document.getElementById("phone").value.trim();
        // Clear old errors
        document.getElementById("name-error").textContent = "";
        document.getElementById("address-error").textContent = "";
        document.getElementById("pincode-error").textContent = "";
        document.getElementById("phone-error").textContent = "";
        let valid = true;
       // Name validation
        if (name === "") {
            document.getElementById("name-error").textContent =
                "Name is required";
            valid = false;
        }
        // Address validation
        if (address === "") {
           document.getElementById("address-error").textContent =
                "Address is required";
            valid = false;
        }
        // Pincode validation
        if (!/^\d{6}$/.test(pincode)) {

            document.getElementById("pincode-error").textContent =
                "Pincode must be exactly 6 digits";
            valid = false;
        }
        // Phone validation
       if (!/^\d{10}$/.test(phone)) {
```

```
            document.getElementById("phone-error").textContent =
                "Phone must be exactly 10 digits";
            valid = false;
        }
        // TASK 8: Successful order
        // ================================================
        if (valid) {
            document.getElementById("confirmation").textContent =
                "Order placed successfully!";
            cart = [];
            localStorage.removeItem("cart");
            updateCartCount();
           form.reset();
        }
    });
}
// RUN ALL FUNCTIONS
// ================================================
document.addEventListener("DOMContentLoaded", function() {

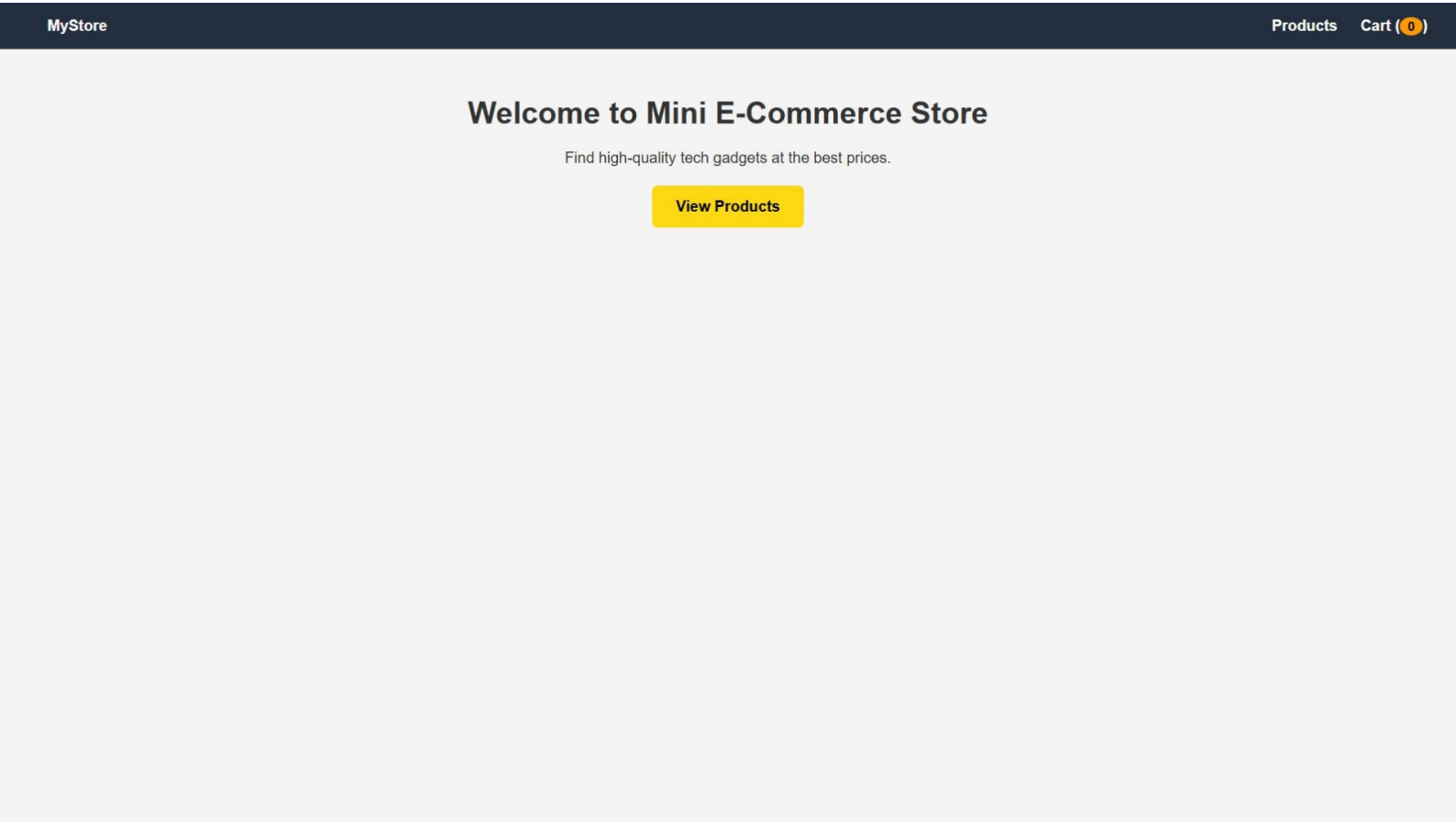
    displayProducts();

    displayCart();

    updateCartCount();

    setupCheckout();

});
```


MyStore
Products
Cart (0)
Welcome to Mini E-Commerce Store
Find high-quality tech gadgets at the best prices.
View Products

MyStore Products Cart (0)

Products Catalog



Wireless Headphones
₹1999
Add to Cart



Smart Watch
₹2999
Add to Cart



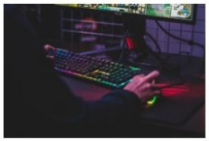
Mechanical Keyboard
₹3499
Add to Cart



Gaming Mouse
₹1299
Add to Cart



Bluetooth Speaker
₹1799
Add to Cart



USB-C Hub
₹899
Add to Cart



Laptop Stand
₹699
Add to Cart

Power Bank 20000mAh
Power Bank 20000mAh
₹1499
Add to Cart

MyStore

127.0.0.1:5500 says

Product added to cart!

OK

Products Cart (0)

Products Catalog



Wireless Headphones

₹1999

Add to Cart



Smart Watch

₹2999

Add to Cart



Mechanical Keyboard

₹3499

Add to Cart



Gaming Mouse

₹1299

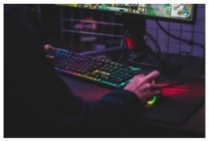
Add to Cart



Bluetooth Speaker

₹1799

Add to Cart



USB-C Hub

₹899

Add to Cart



Laptop Stand

₹699

Add to Cart

Power Bank 20000mAh

Power Bank 20000mAh

₹1499

Add to Cart

MyStore Products Cart (1)

Shopping Cart

	Wireless Headphones	₹1999	1	₹1999	Remove

Total: ₹1999

Proceed to Checkout

Checkout

Full Name

Delivery Address

Pincode (6 digits)

Phone Number (10 digits)

Place Order